Automated College Timetable Scheduler

Feature Showcase Booklet

Combinatorial & Spatial Optimization

This platform addresses the notoriously **NP-hard University Course Timetabling Problem (UCTP)** by coupling an industrial-strength **Constraint Programming (Google OR-Tools CP-SAT)** solver with a visual **2D Campus Spatial Graph Engine**.

Beyond satisfying standard curricular requirements, teacher workloads, and room capacities, the system incorporates exact walking-distance models across multi-floor campus facilities, actively eliminating transit fatigue between back-to-back lecture periods.

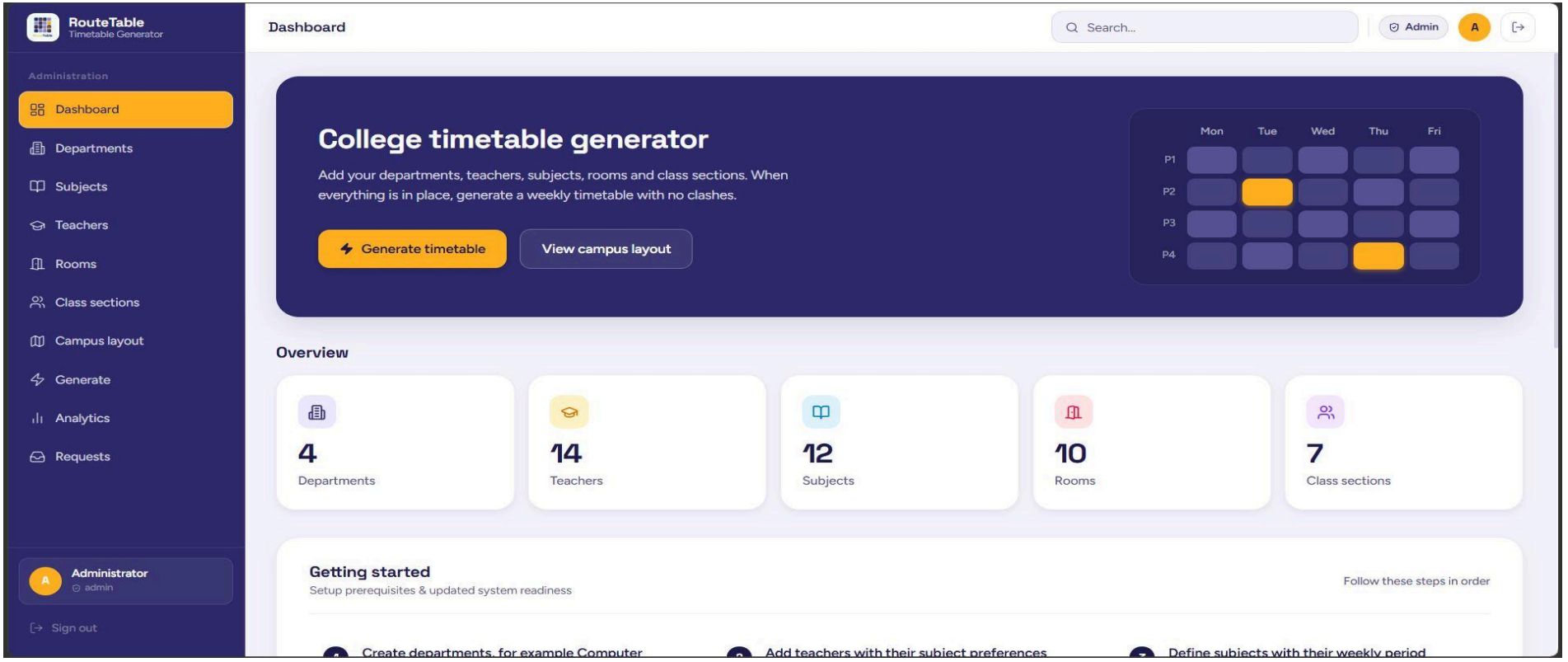
Core Architectural Pillars

Google OR-Tools CP-SAT: Mathematical hard constraints (C1–C6) and multi-objective soft penalties (O1–O3).
2D Floor Grid Painter: Cell-level spatial modeling with automated room blob detection and BFS transit pathfinding.
Continuous Lab Blocks: Automated packing of multi-period practical sessions into uninterrupted schedules.
Differential Slot Locking: Fix approved slots while incrementally re-solving remaining assignments.
Full Academic Analytics: Room utilization heatmaps, workload balance metrics, and transit distance auditing.

Backend	Constraint Solver	Frontend UI	Database & Storage	Document Engine
FastAPI 0.115 • Python 3.11+	Google OR-Tools CP-SAT	React 18 • Vite • Tailwind	PostgreSQL 16 • SQLAlchemy	ReportLab • OpenPyXL

Executive Administration Dashboard

The central operational command center provides real-time academic resource counts, active timetable statuses, quick-launch shortcuts, and live solver telemetry.



Core Statistics

Monitors institutional totals: Departments, Faculty, Classrooms, Specialized Labs, and Active Cohorts at a glance.

Quick Action Hub

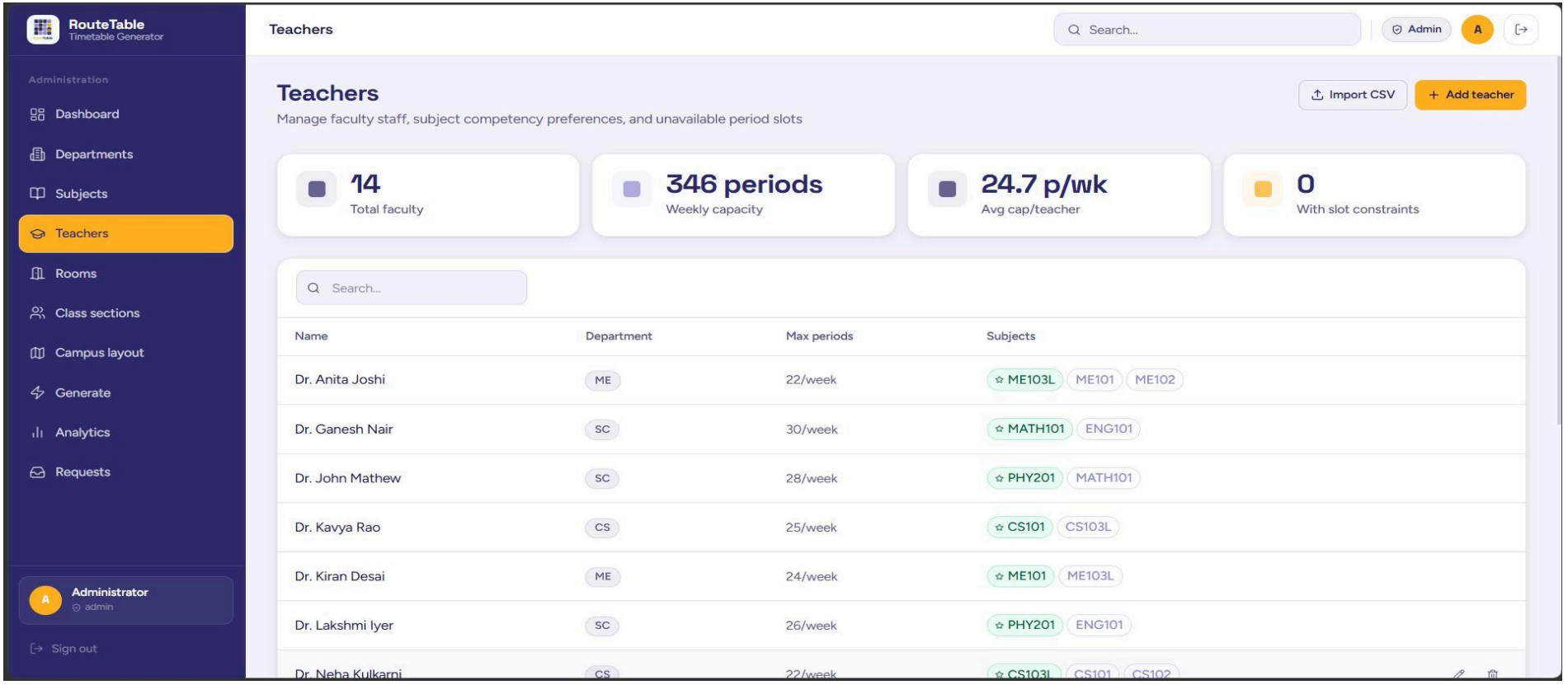
Provides one-click navigation to Timetable Generator, Campus Grid Painter, and Academic Setup modules.

Run Auditing

Displays recent generation job logs, active status tags (Optimized / Feasible), and wall-time execution metrics.

Faculty Directory & Workload Profiling

Comprehensive faculty profile management detailing departmental affiliations, maximum weekly teaching period caps, and subject competency mappings.



Workload Caps

Enforces strict contractual limits (e.g. max 25 periods/week) via hard constraint C6 during schedule generation.

Preference Affinity

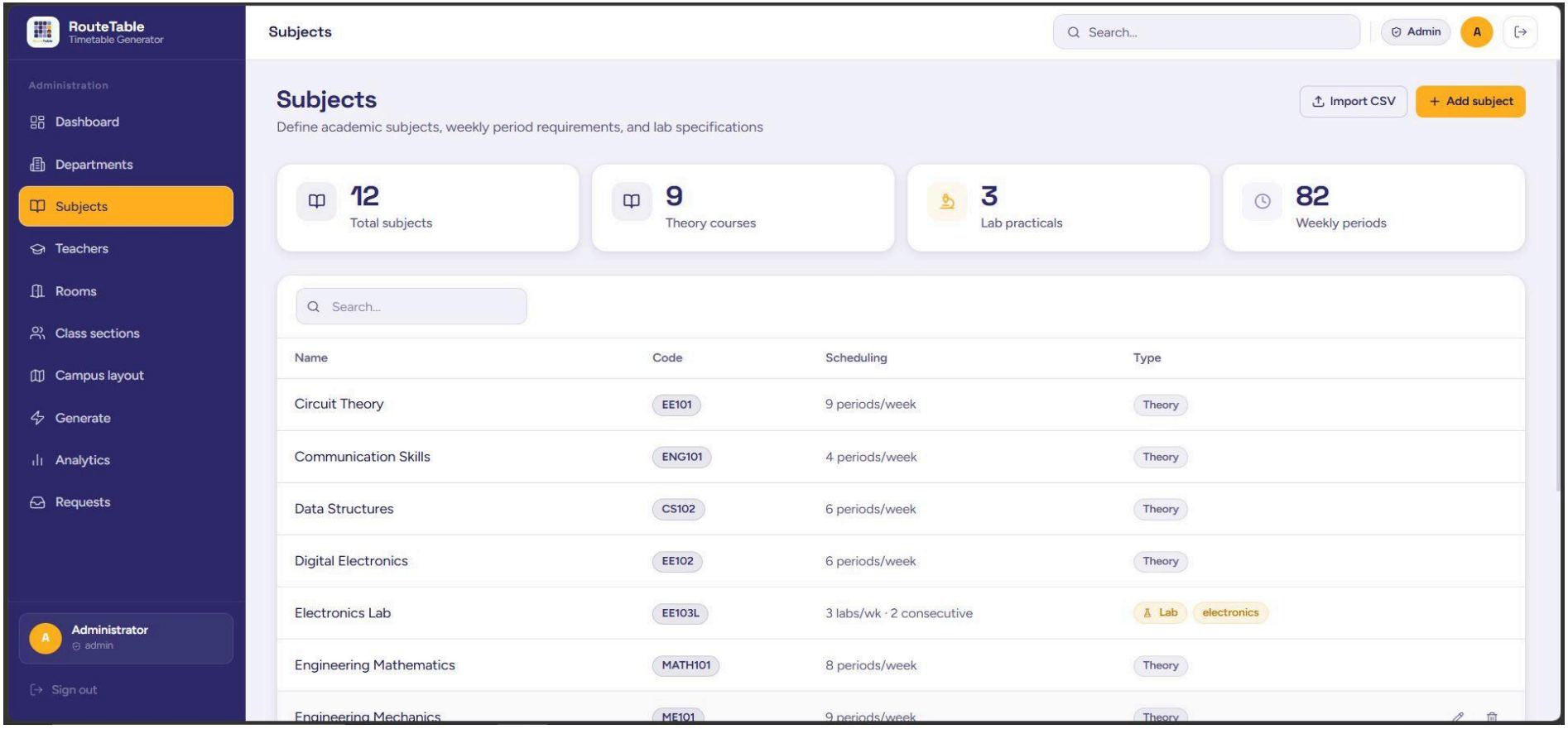
Maps teacher subject preferences across Preferred, Willing, and Avoided categories with priority ranks 1 to 3.

Blackout Windows

Configures teacher day/period unavailability slots

Course Catalog & Laboratory Configurations

Academic curriculum repository categorizing lecture courses and specialized multi-period laboratory practical sessions.



Curriculum Periods

Defines weekly required lecture period counts for every academic subject with section-level override options.

Multi-Period Blocks

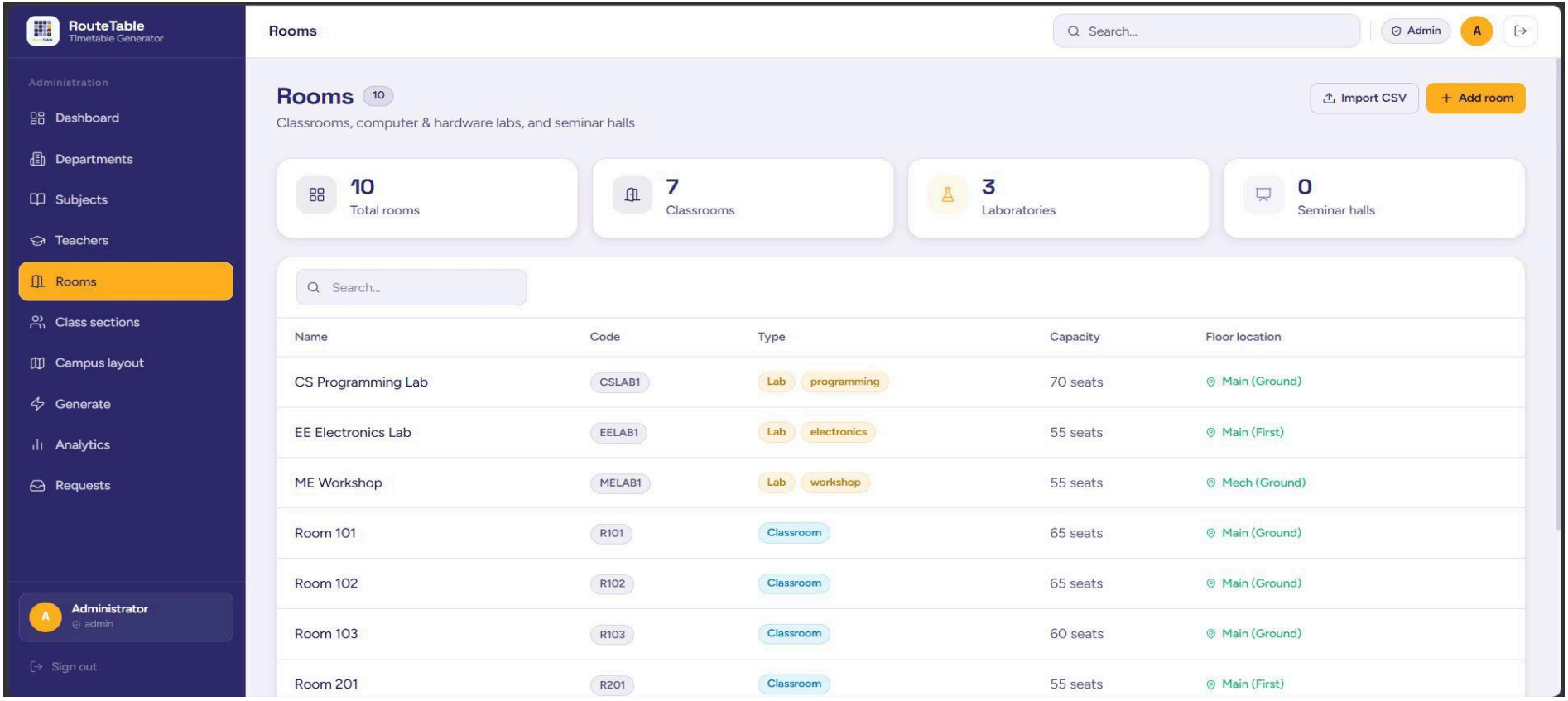
Specifies laboratory session requirements (e.g., 2 or 3 continuous periods) packed seamlessly by the solver.

Hardware Matching

Binds subjects to required lab facilities (Computer, Electronics, Chemistry) to enforce room suitability.

Classroom & Specialized Lab Facilities

Facility asset management registering lecture halls, computer centers, and specialized laboratories with capacity tracking and building locations.



Room Classification

Distinguishes standard classrooms, large seminar auditoriums, and specialized technical laboratories.

Capacity Verification

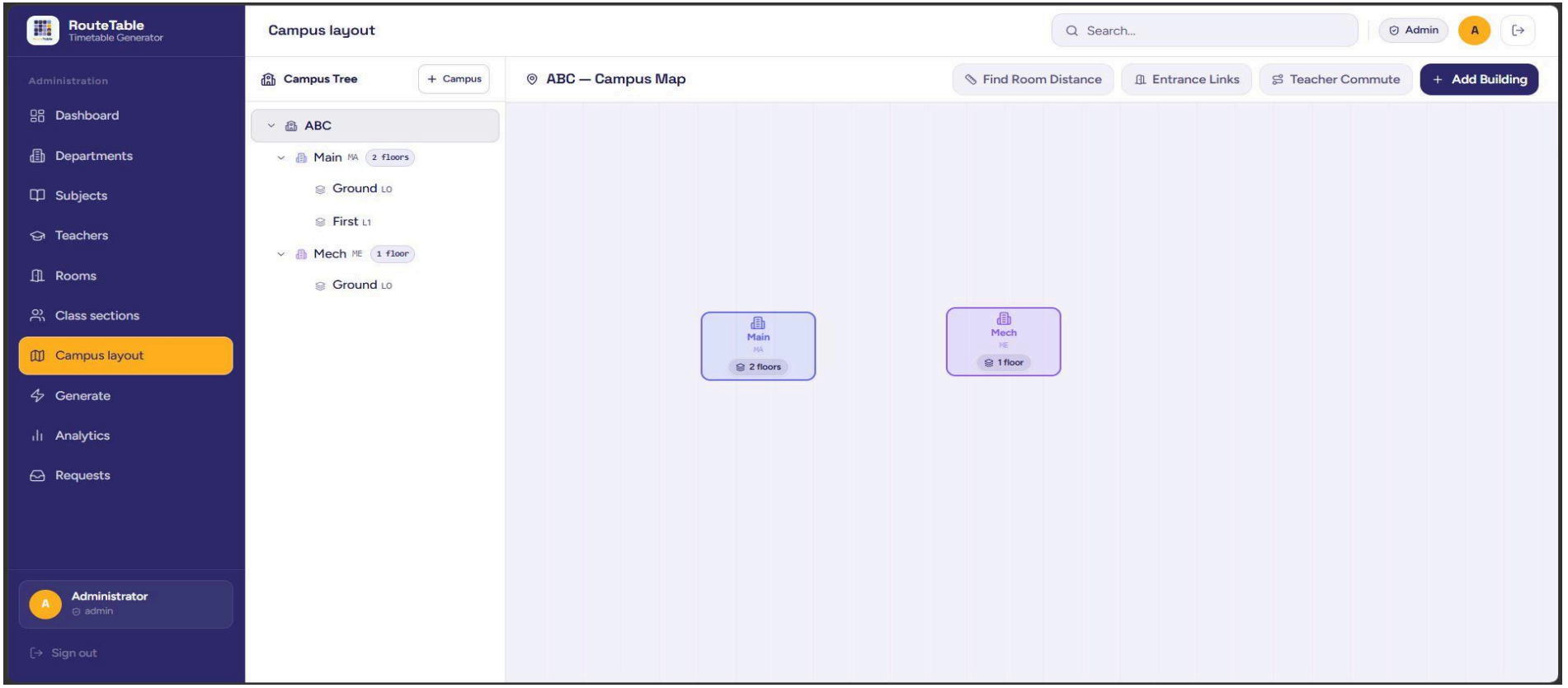
Tracks seat capacity to ensure assigned student cohort strength never exceeds available room capacity.

Spatial Association

Links each room to a physical building and floor level for shortest-path walking transit calculations.

Interactive Campus Geographic Overview

High-level visual canvas of the collegiate campus displaying building placements, 2D coordinates, footprint dimensions, and outdoor walkway connections.



Visual Building Blocks

Customizable building bounding boxes, spatial coordinates (X, Y), and identifying departmental color palettes.

Inter-Facility Networks

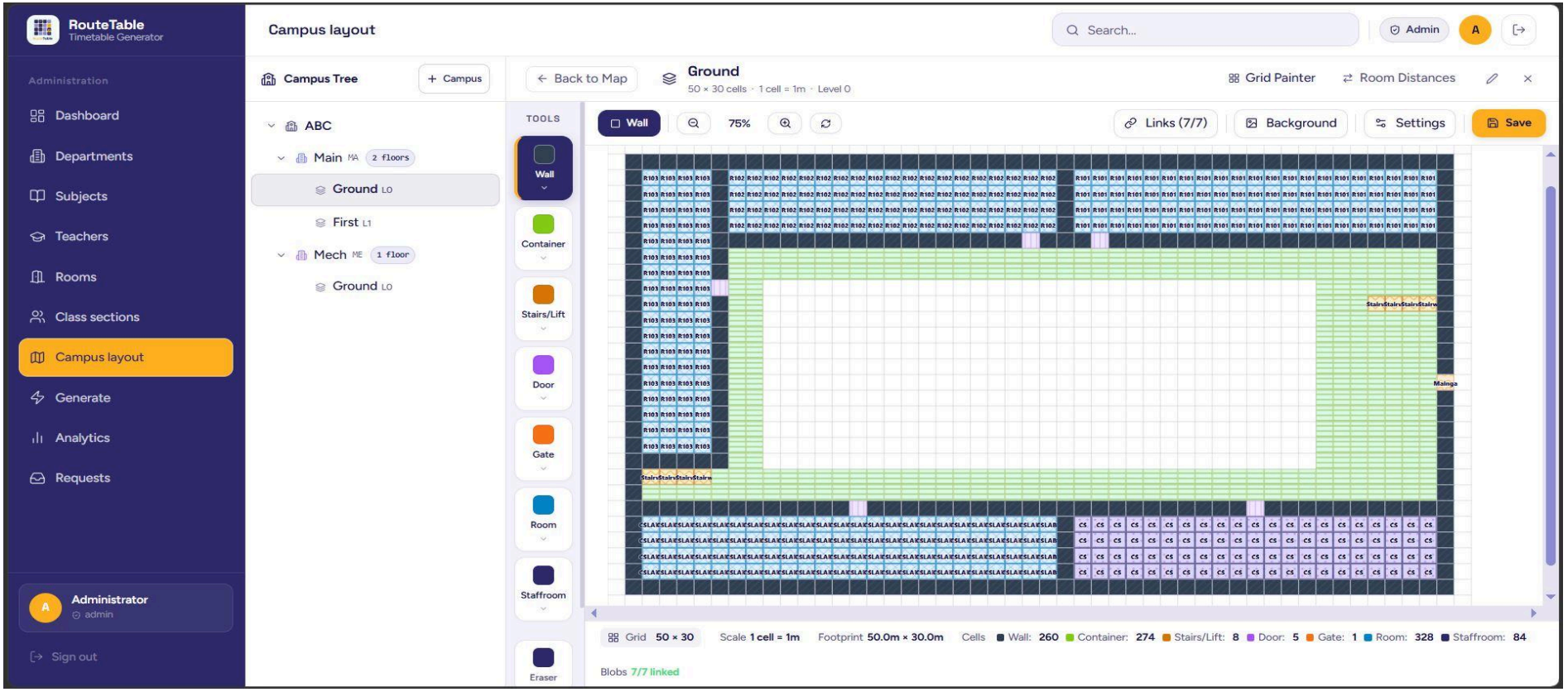
Defines walking routes, perimeter gates, and transit corridors connecting disparate campus buildings.

Hierarchical Model

Establishes the master physical tree: Campus → Buildings →Multi-level Floors → Facility Blobs.

2D Interactive Floor Grid Painter & BFS Transit Router

Browser-based 2D grid architectural designer allowing administrators to paint walkable corridors, doors, stairs, gates, and link them directly to room entities.



Grid Matrix Painting

Paint corridors, walls, doors, stairs, and gates at 1m scale directly in the browser with brush tools.

Entity Association

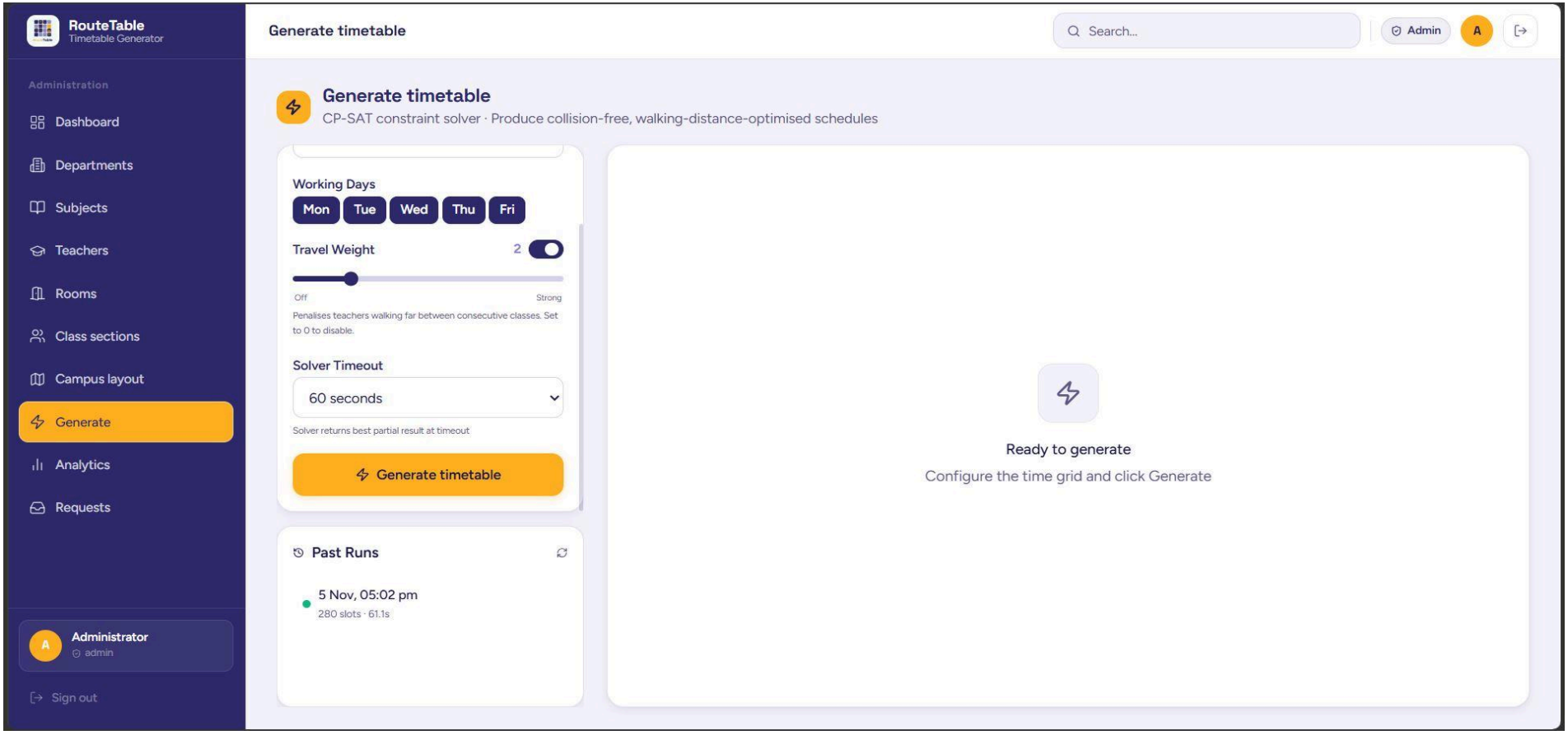
Automatically detects painted room cell blobs and links them to database room and staff room records.

Shortest-Path BFS

Computes exact intra-floor and cross-floor walking transit distances used by the solver to minimize walking fatigue.

Timetable Generation Engine & Solver Controller

Execution control panel parameterizing the Google OR-Tools CP-SAT solver, objective weight multipliers, and mathematical search bounds.



Time Grid Setup

Configures institutional calendar rules: working days per week (5–6 days) and daily period counts (6–8 periods).

Objective Balancing

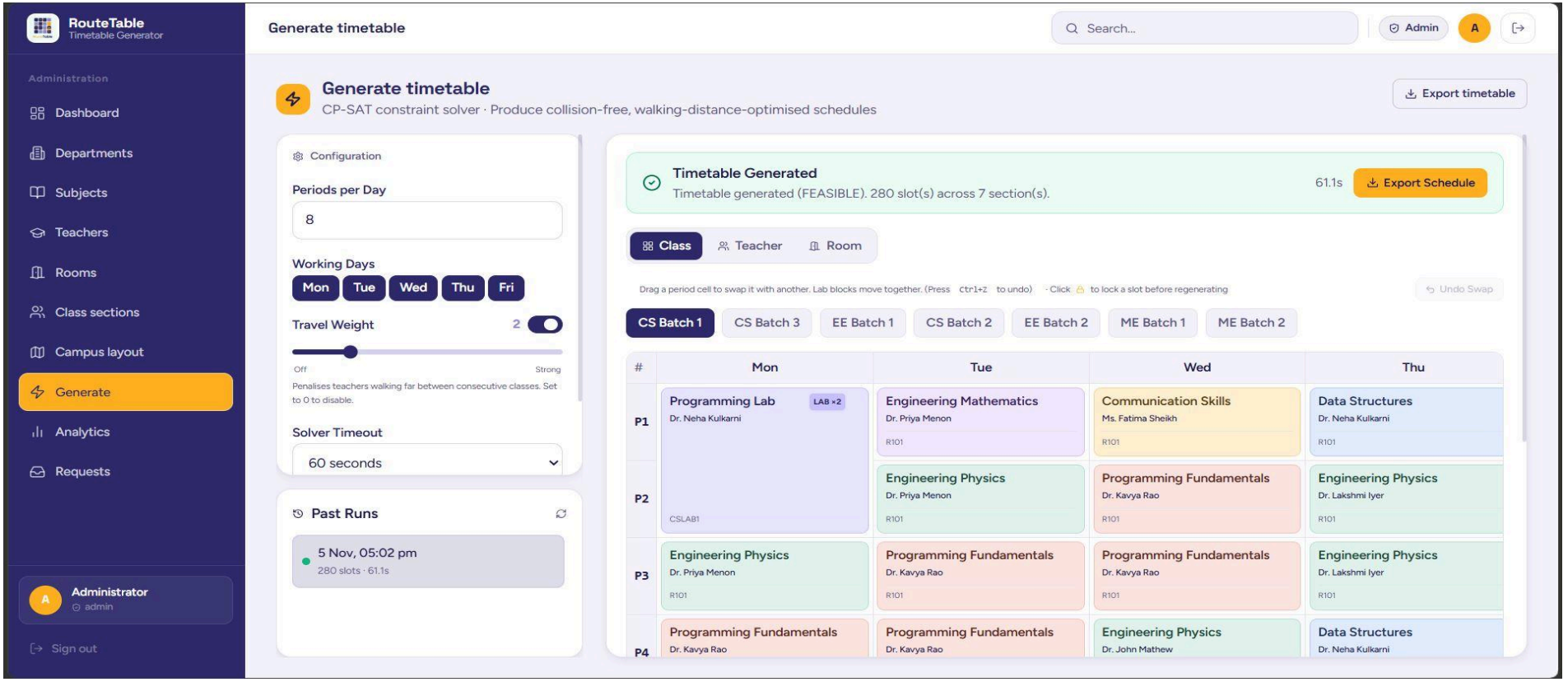
Fine-tunes solver priorities: Weekly Spread Weight, Teacher Preference Weight, and Walking Distance Penalty.

Real-Time Telemetry

Monitors branch-and-bound search, constraint satisfaction wall-time, and optimality gap metrics.

Master Timetable Matrix & Interactive Slot Locking

Conflict-free schedule matrix displaying theory lectures, unbroken laboratory practical blocks, and interactive slot locking controls.



Dynamic Perspective

Toggles views seamlessly between Class Section schedules, Teacher weekly assignments, and Room occupancies.

Continuous Lab Packing

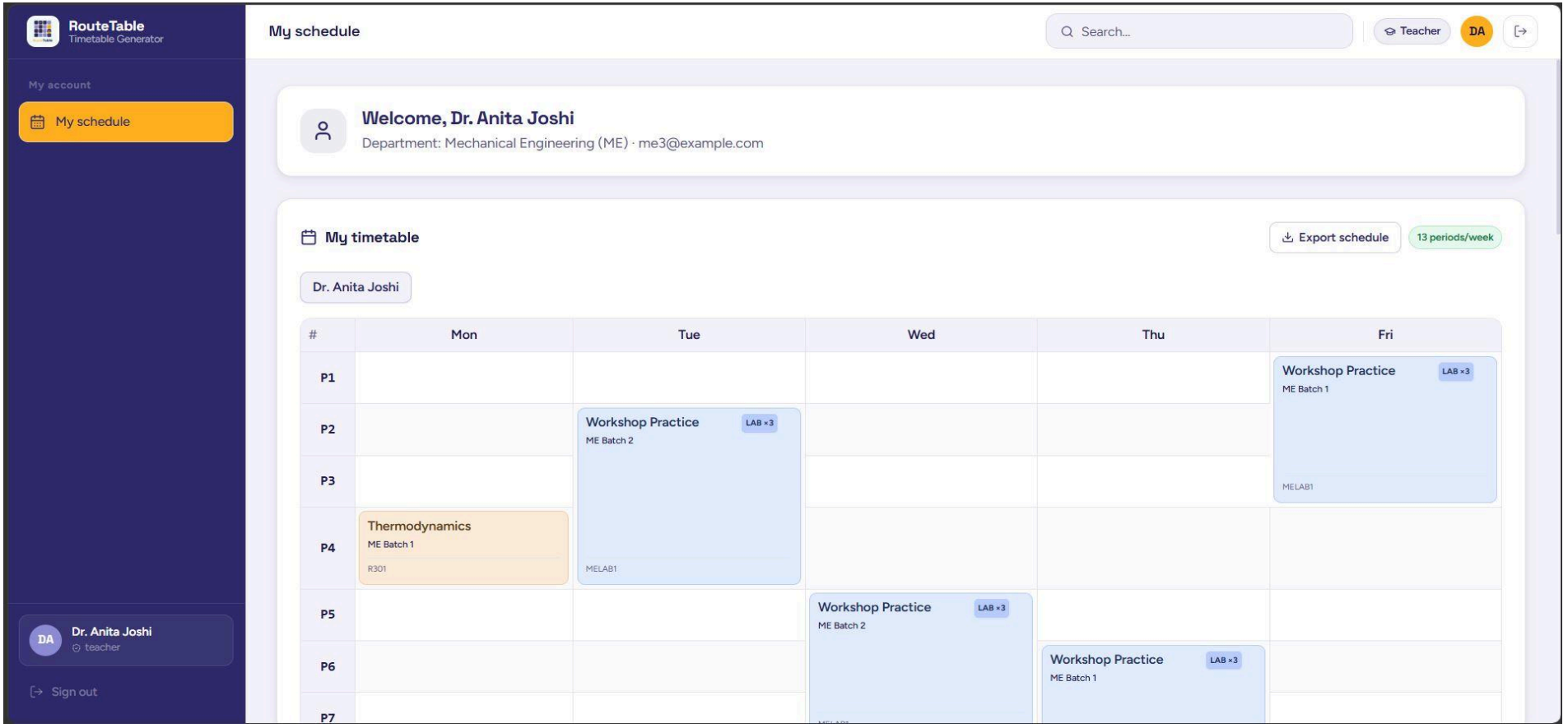
Renders multi-period lab blocks spanning consecutive slots without subject or instructor interruptions.

Slot Locking

Interactive padlock control fixes approved slots in place, enabling differential re-solves around fixed commitments.

Teacher Self-Service Portal & Substitution Workflow

Dedicated instructor interface presenting personal weekly schedules, room-to-room transit instructions, and substitution request workflows.



Instructor Dashboard

Shows daily teaching agenda, assigned student batches, and room numbers formatted for desktop and mobile view.

Transit Guidance

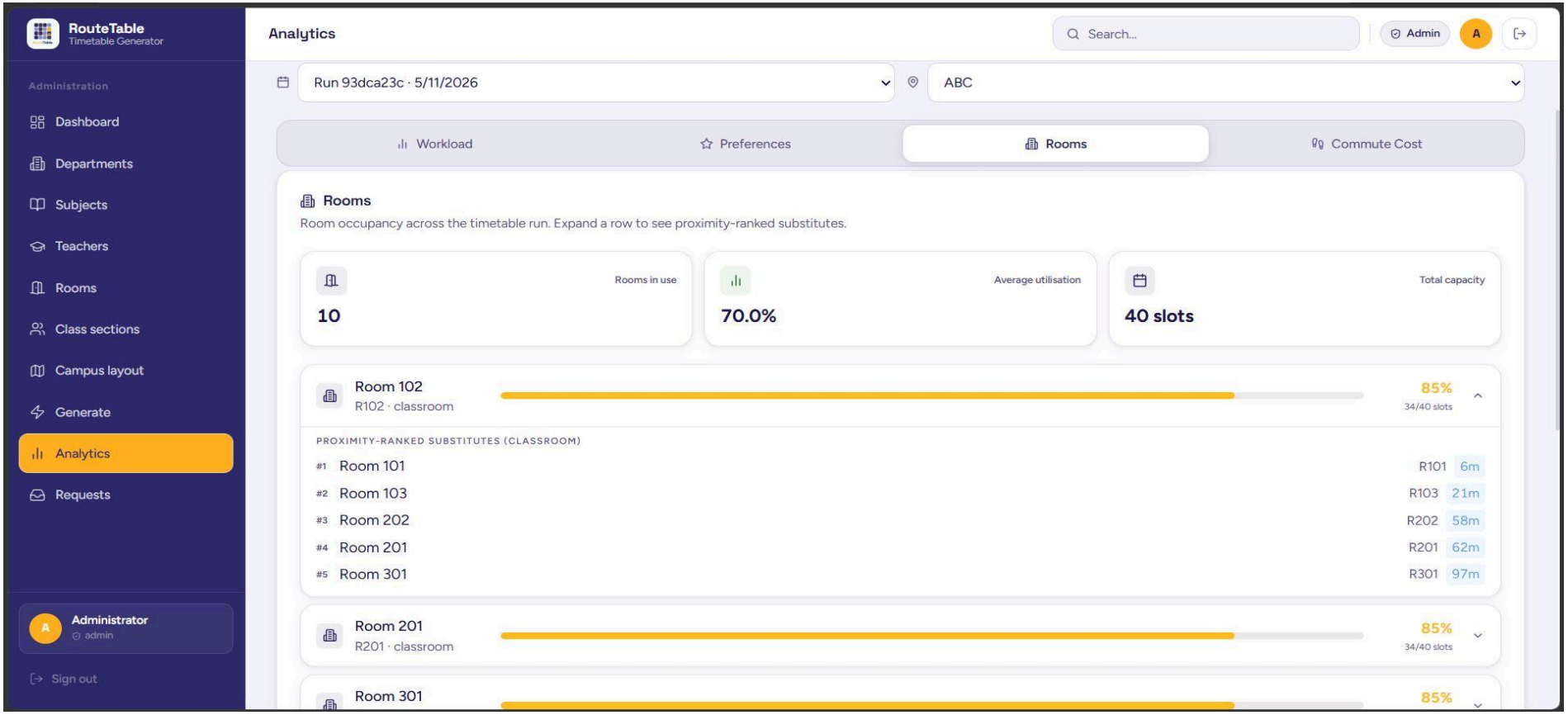
Displays destination room locations, floor levels, and recommended walking paths between consecutive periods.

Substitution Pipeline

Allows teachers to initiate schedule adjustment or leave-coverage requests directly to department heads.

Facility Utilization & Capacity Heatmaps

Interactive visual heatmap tracking classroom and laboratory occupancy rates across all academic periods and weekdays.



Utilization Percentages

Computes exact percentage utilization for each lecture hall and laboratory across the academic week.

Visual Heatmap Matrix

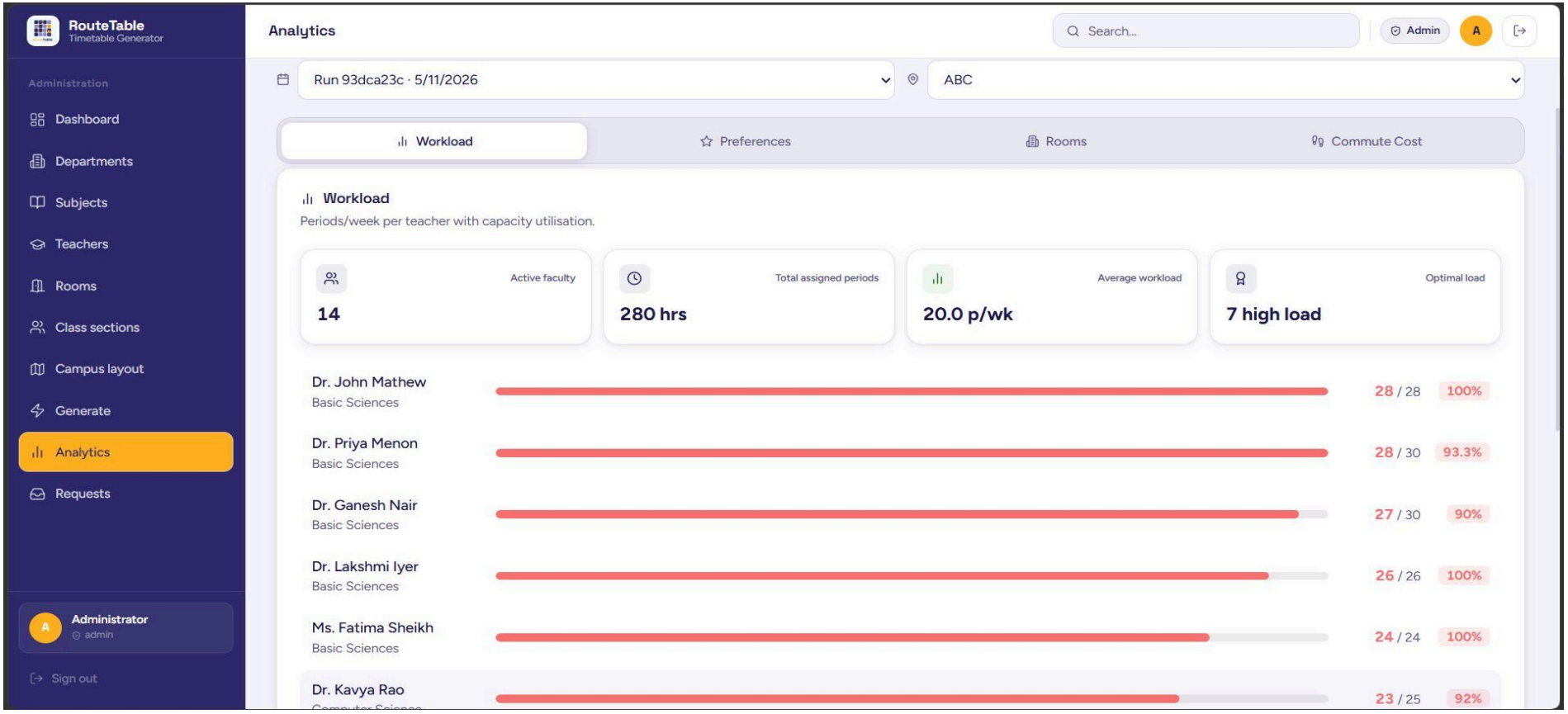
Color-coded intensity grid immediately reveals high-demand bottleneck hours and underutilized slots.

Capital Optimization

Aids academic leadership in making evidence-based decisions on facility sharing and lab investments.

Faculty Teaching Workload Distribution

Interactive visual analytics illustrating weekly assigned teaching periods across faculty to guarantee equitable distribution.



Workload Comparison

Bar charts depict weekly teaching periods per instructor grouped by department for transparent oversight.

Cap Compliance

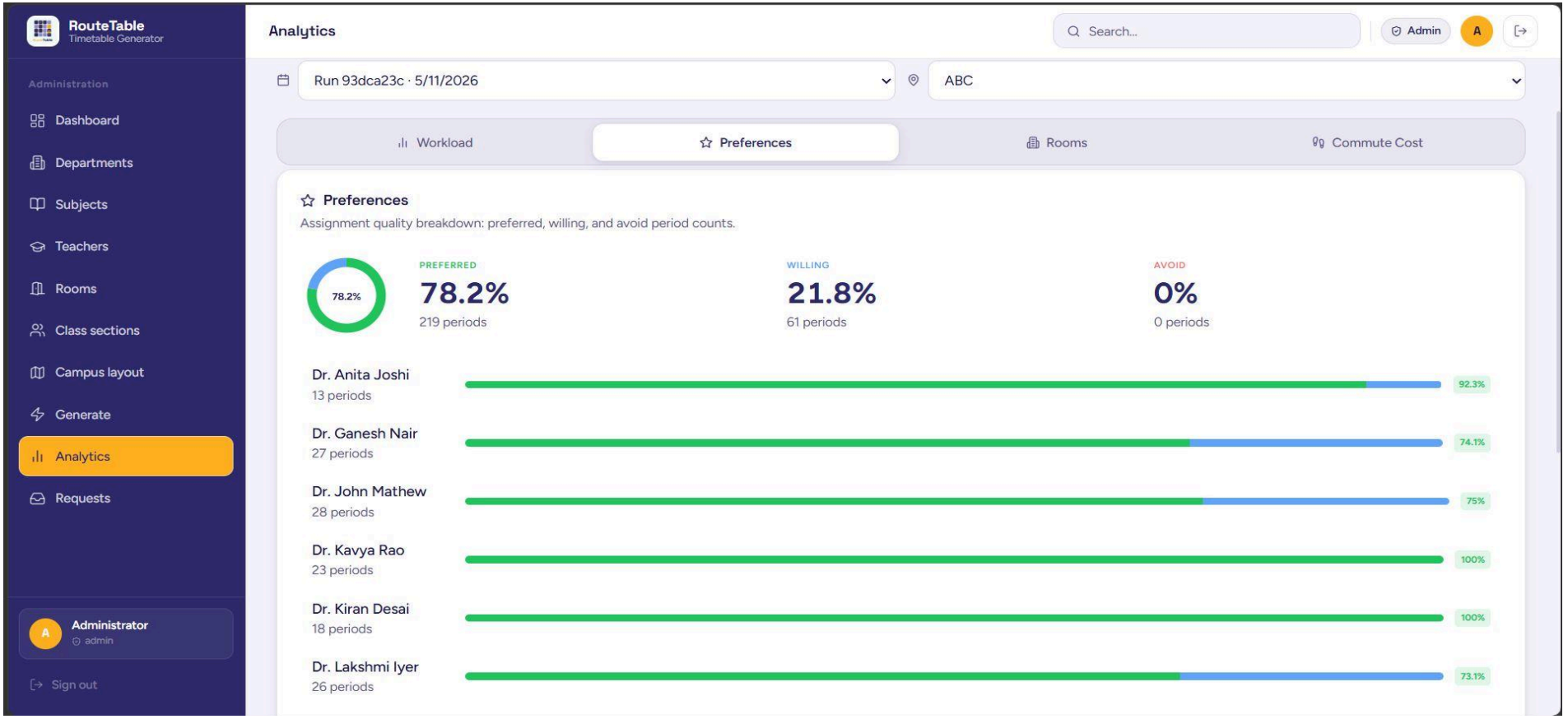
Visually indicates instructors near or at their contractual maximum period limit.

Burnout Prevention

Ensures teaching duties are uniformly distributed across senior professors and junior lecturers.

Faculty Subject Preference Satisfaction

Quantitative metrics measuring how effectively the CP-SAT optimization engine satisfied instructors' subject preference rankings.



Affinity Breakdown

Categorizes total assigned teaching periods across Preferred, Willing, and Avoided classifications.

Priority Satisfaction

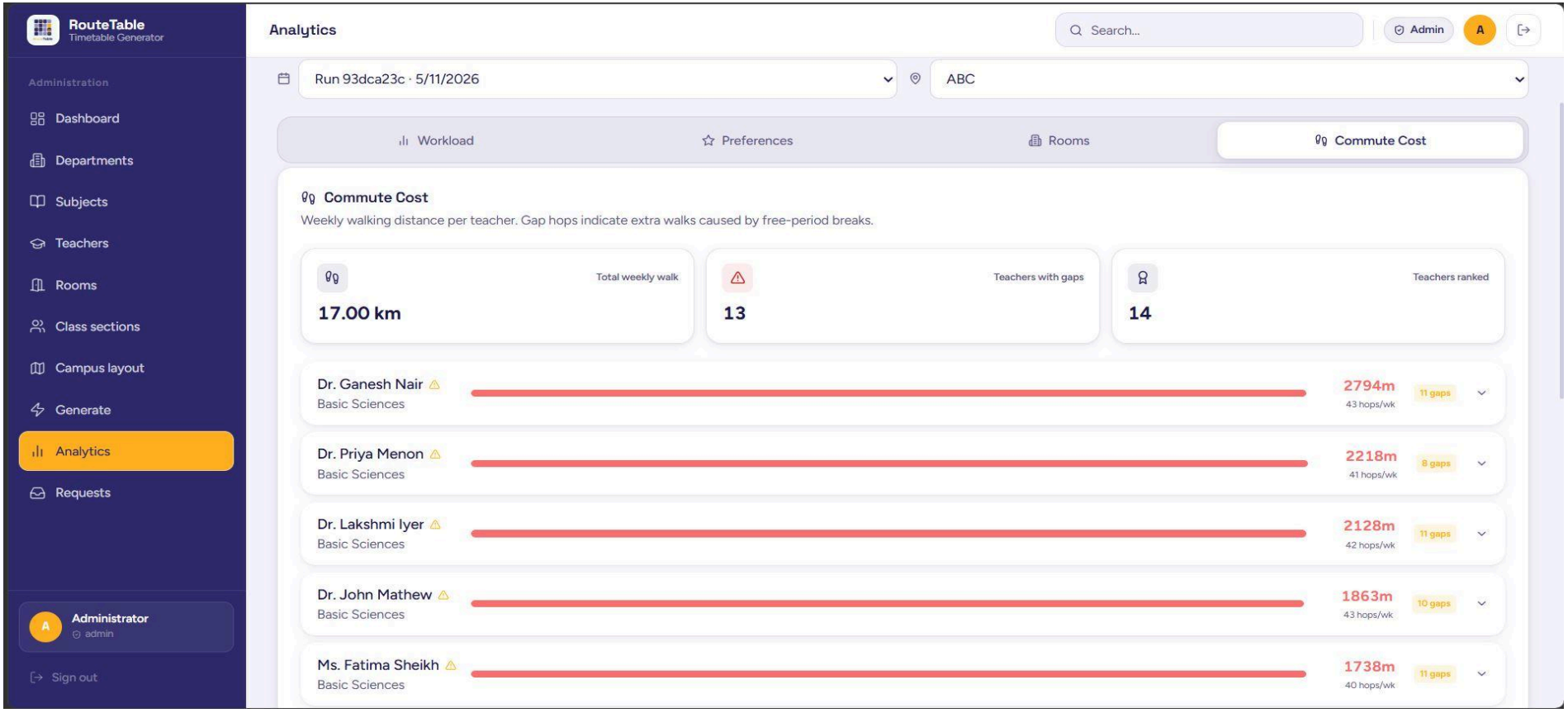
Tracks percentage of instructors receiving their #1 and #2 ranked course preferences.

Penalty Minimization

Demonstrates the efficacy of soft objective in eliminating undesirable course allocations.

Campus Walking Transit & Distance Reduction

Spatial analytics reporting physical walking transit distances traversed by instructors between consecutive class periods.



Transit Metrics

Quantifies total and average daily walking distance (in meters) traversed by instructors across campus.

Vertical Transition Tracking

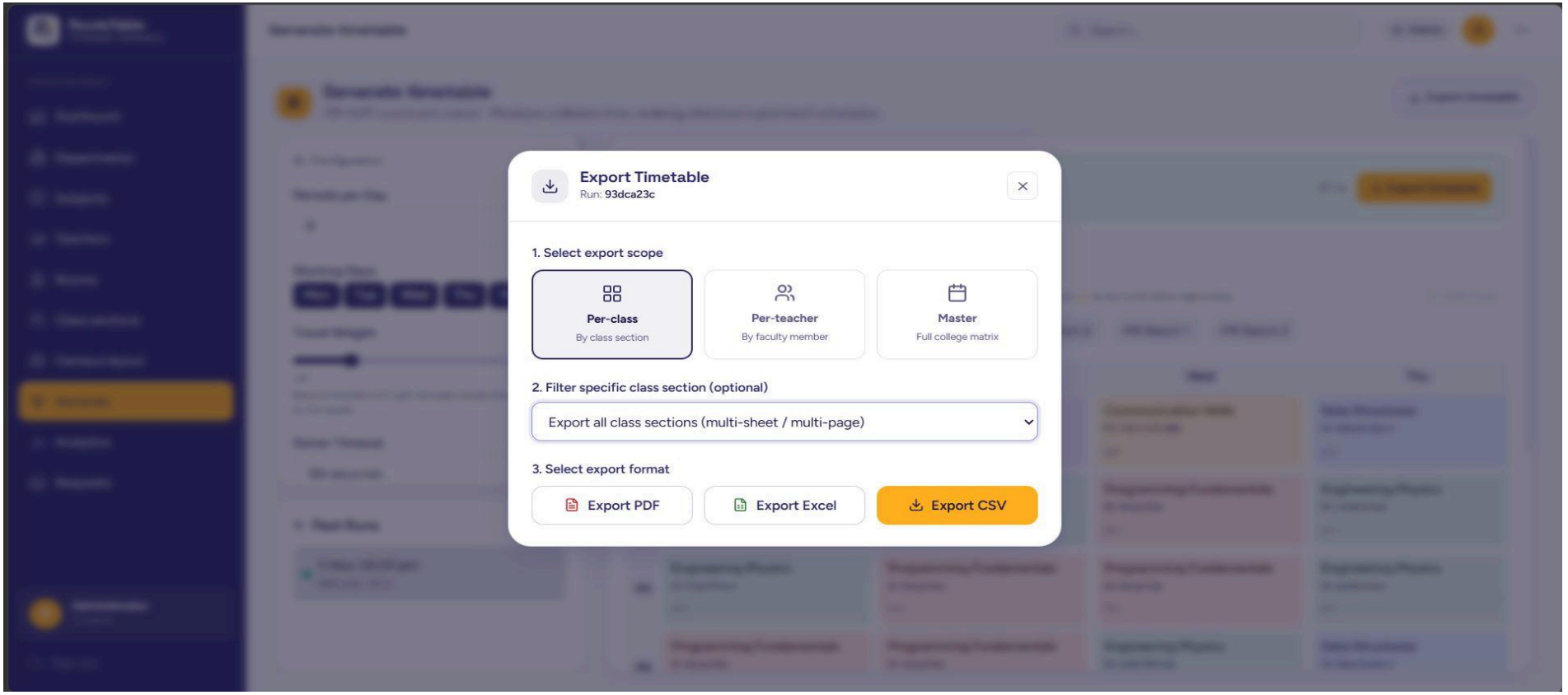
Analyzes stair and elevator crossings between consecutive periods to minimize vertical fatigue.

Spatial Optimization Audit

Validates that soft objective successfully grouped consecutive classes in adjacent rooms or floors.

Multi-Format Export & Enterprise Data Pipeline

Production-ready reporting facilities delivering styled Excel workbooks, print-ready PDF schedules, and bulk CSV ingestion tools.



Multi-Sheet Excel (.xlsx)

Generates styled, color-coded workbooks with separate tabs for Campus Master, Departments, and Sections.

Print-Ready PDF

Produces publication-quality landscape timetable matrices with headers, legends, and department stamps.

Bulk CSV Ingestion

Streamlines onboarding through pre-validated CSV uploads for teachers, subjects, rooms, and sections.